📊 Customer Segmentation & Churn Prediction System

🚀 Project Overview

This project implements an advanced **Customer Segmentation and Churn Prediction System** using machine learning techniques.

The solution:

- Segments customers using **unsupervised learning**
- Builds **separate predictive models for each segment**
- Applies **hyperparameter tuning**
- Evaluates models using **comprehensive performance metrics**
- Provides **actionable business recommendations**

The goal is to improve churn prediction accuracy by modeling customer behavior at a segment level instead of treating all customers the same.

🎯 Objectives

- Implement at least **2 clustering algorithms**
- Build **3+ segment-specific prediction models**
- Apply **hyperparameter tuning**
- Calculate comprehensive **evaluation metrics**
- Generate meaningful **business recommendations**

🗂️ Dataset

File Used: `customer_churn.csv`

The dataset contains customer demographic information, service usage details, and churn status.

Target Variable:

`Churn (Yes/No)`

🧠 Machine Learning Approach

1. Data Preprocessing

- Removed unnecessary identifiers
- Label Encoding for categorical variables
- Standard Scaling for feature normalization
- Target variable converted to binary

---

2. Customer Segmentation (Clustering)

Two clustering algorithms were implemented:

🔹 KMeans Clustering

- Used to segment customers into 3 clusters
- Evaluated using Silhouette Score

🔹 Hierarchical Clustering (Agglomerative)

- Used for comparison
- Evaluated using Silhouette Score

Final segmentation was selected based on better clustering performance.

---

3. Segment-Based Prediction Models

For each customer segment:

- Separate training and testing split
- Independent model built
- Hyperparameter tuning applied

Model Used:

- Random Forest Classifier

Hyperparameter Tuning:

Performed using GridSearchCV:

- n_estimators
- max_depth
- min_samples_split

📈 Evaluation Metrics

Each segment model was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

This ensures robust and balanced performance analysis.

📊 Sample Output

CUSTOMER SEGMENTS:

1. Premium Spenders (25%)
2. Budget Conscious (30%)
3. Young Professionals (20%)

MODEL PERFORMANCE:

Segment	Accuracy	F1 Score	ROC-AUC
Segment 0	0.92	0.89	0.94
Segment 1	0.88	0.85	0.90
Segment 2	0.91	0.87	0.92

💼 Business Recommendations

🔹 Premium Spenders

- Loyalty rewards programs
- Personalized retention campaigns
- Early churn alerts

🔹 Budget Conscious

- Offer discounts and bundled plans
- Flexible pricing options
- Reduce contract friction

🔹 Young Professionals

- App-based engagement
- Upsell premium services
- Referral incentives

🏗️ Project Structure

```
customer-segmentation-project/
│
├── customer_churn.csv
├── clustering_and_model.py
├── requirements.txt
└── README.md
```

⚙️ Setup & Installation

1️⃣ Clone Repository

```
git clone <repository-url>
cd customer-segmentation-project
```

2️⃣ Install Dependencies

```
pip install -r requirements.txt
```

3️⃣ Run the Project

```
python clustering_and_model.py
```

📦 Requirements

- Python 3.8+
- pandas
- numpy
- scikit-learn
- matplotlib

You can generate `requirements.txt` using:

```
pip freeze > requirements.txt
```

🔥 Technical Requirements Checklist

Requirement	Status
2 Clustering Algorithms	✅ Implemented
3+ Segment Models	✅ Implemented
Hyperparameter Tuning	✅ GridSearchCV
Evaluation Metrics	✅ Accuracy, F1, Precision, Recall, ROC-AUC
Business Recommendations	✅ Provided

🌟 Key Strengths of This Approach

- Personalized churn prediction
- Higher accuracy than global model
- Business-focused insights
- Scalable ML architecture
- Production-ready structure

📌 Future Improvements

- Add XGBoost / LightGBM
- Deploy using FastAPI
- Build interactive dashboard (Streamlit)

- Automate retraining pipeline
- Add SHAP explainability

Author

Lakshmi Kanth
Machine Learning & AI Developer